



Purple Team

FA2026 • 2026-10-06

CCDC: SOC Overview

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Pranav Srinivasan

- SigPwny Purple Team Lead
- Computer Engineering '28
- Fun fact: Can kinda rival Ronan's cursed course credit history



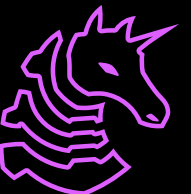
Michael Khalaf

- SIGPwny: Admin, Purple Team
Cyber Defense Lead, Purple Lead
(2024-2026)
- MSIM & ICSSP Class of May 2026
- NetSec (NetOps, SysAdmin), DFIR,
Pen, OSINT



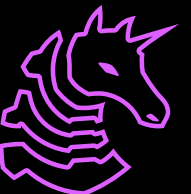
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 - a. Palo Alto NGFW
 - b. Cisco FTD (Firepower Threat Defense)
6. CCDC



What is System Hardening

- Process of securing a computer system by reducing its vulnerable areas and closing off entry points for attackers
 - IMPLEMENTING (scripting) ZERO-TRUST ARCHITECTURE
-
1. Understanding the system
 2. Rotate passwords
 3. Identify and delete unused or unnecessary accounts, services, processes, apps, etc.
 4. Lock down the network
 5. Enable SIEM
 6. Update Systems to latest version



What is Incident Response?

- Structured process of identifying, managing, and mitigating the effects of cybersecurity incidents to minimize damage, recover operations, and prevent future occurrences
 - Depends on organization, but priority not just cleanest version, but also uptime (SLA violations), business continuity, compliance, evidence preservation, etc. also important
1. System Identification: identify the affected systems.
 2. System Isolation: isolate compromised systems from the network.
 3. System Investigation: investigate what happened and how.
 4. System Recovery: restore systems to a safe, working state.
 5. System Monitoring: monitor systems for further suspicious activity.



CCDC: Collegiate Cyber Defense Competition

- CCDC is a blue team only competition for university students.
- Teams inherit a virtual network and are tasked with **defending it in real time** against the CCDC red team
- Must **maintain uptime** of critical business services, more uptime means more successful automated scoring (web, email, DNS, etc)
- Must respond to **injects** (business tasks) in the form of management memos (policies, reports, config changes) under **tight deadlines**
- **Incident Response** reports must also be submitted once adversary activity is discovered (with remediations)
- **Structure: Invitationals → Qualifiers → Regionals → Nationals**



CCDC National Website (new)



<https://www.ccdc.io/>

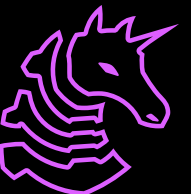


CCDC: The Classic Feel



Automation

- Lot of these capabilities can be automated and scripted
 - Dev Work will be done to prepare scripts that can be run on the systems to immediately harden and deploy monitoring
 - In previous years, scripts written for a “league-altering” AI-EDR system.
-
- However, a big part of the competition is documentation and report writing (Injects), we don’t script or automate these
 - In security, we say RTFM, but we also ask to not AI-slop the Docs. Good practice for professional report writing anyway. (I will boot you from CCDC if you AI-slop docs)



Automation: Scripting

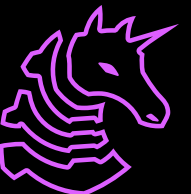
- Scripting in bash and powershell is extremely important to roll out configs and changes to the endpoints you're defending.
- BASH, Powershell

Useful tools: John Ford CCDC Defender Toolkit, Ansible



Security Information & Event Management

- A SIEM (Security Information and Event Management) system pulls logs from across the environment, including firewalls, servers, endpoints and applications into one searchable place
- It normalizes the data and runs correlation rules to spot patterns that one log source alone would miss.
- When a rule matches, it raises an alert. It's designed to give you a holistic view of scattered system activities hard to detect alone. Ingests these log sources, aggregates all of these logs, and offers telemetry with more granular insight
 - You can query events by endpoint, date/time, behavior, services
 - You will need this for confirming configs, baseline behaviors, DFIR, and threat hunting



EDR: Endpoint Detection & Response

Endpoint security

Endpoint detections

Search

Detections 23 results (243 total)

Search detections

Severity

Time

Status

Tactic

Technique

Tags

Command line

Grouping tags

Add/remove filters +

Clear all

List is up to date

Aggregate detections Off

Group by

Sort by Time: Newest to oldest

Severity

Detect time

Name

Attributes

Aug. 12, 2025

Severity

High

Detect time

09:00:44

Process on host

powershell.exe on SRV-01 by Administrator

Tactic via technique

Defense Evasion via

Command line

"C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe" -WIN...

Aug. 10, 2025

Severity

High

Detect time

15:34:51

Process on host

Setup.exe on LPT-12766 by j.adams

Tactic via technique

Machine Learning ...

Command line

"C:\Users\j.adams\Downloads\Photoshop 26.3 (Cracked)\Setup.exe"

Severity

Low

Detect time

14:01:45

Process on host

cmd.exe on LPT-10075 by k.smith

Tactic via technique

Malware via Adware

Command line

"C:\WINDOWS\system32\cmd.exe" /d /c TIMEOUT 5 & cmd /d /c del "C:...

Severity

Low

Detect time

14:01:45

Process on host

appbundle.exe on LPT-10075 by k.smith

Tactic via technique

Malware via Adware

Triggering file

appbundle.exe

Hostname

LPT-10075

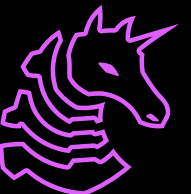
User name

k.smith



EDR: Endpoint Detection & Response

- EDR (Endpoint Detection and Response) monitors activity on endpoints like processes, file changes and network connections
- It catches threats by their behavior rather than just known signatures.
- When it finds these, we can investigate the attack timeline and respond remotely, for example by isolating a host or killing a process.



Splunk

Why We Use It

- Central Log Hub: Aggregates and normalizes scattered logs across all operating systems, endpoints, next-gen firewalls, and network subnets into a single searchable location.
- Fast Incident Response: Provides rapid querying of behaviors, endpoints, and specific dates/times, allowing the team to trace an attack timeline and confirm system hardening configs during an active incident.

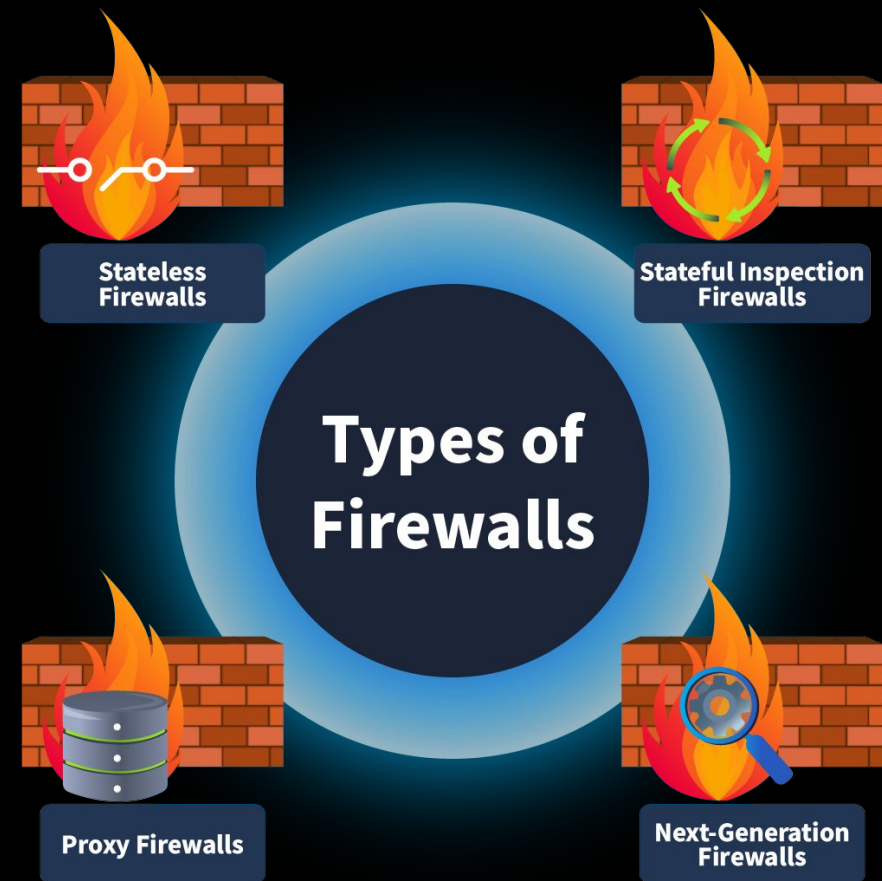
How It Is Deployed

- Agent-Based Ingestion: Deployed by installing a lightweight Splunk Universal Forwarder (UF) agent onto individual endpoints to securely ship system, application, and network event logs to the central indexing engine.



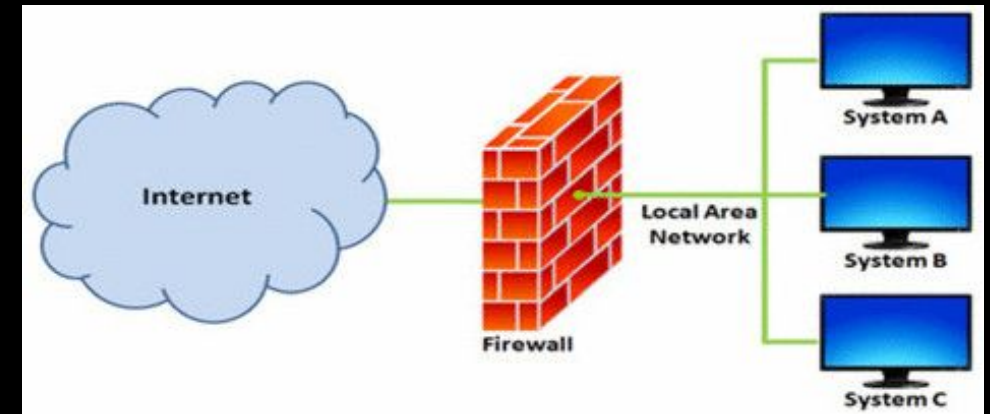
Firewall Overview

- Monitors incoming and outgoing network traffic to look for any suspicious activities
- Proxy firewalls operate on layer 7
- Next-gen firewalls operate on levels 3 and 7 with heuristic analysis



What is a Firewall

- When asking the question, “how does a blue teamer defend against malicious traffic in a live setting” there is one reliable option for **live detection**
- Firewalls are the technical utility to achieve this
- They are locks on doors between (hopefully) segmented networks (in many cases, VLANs)
- They are both GUI and command line accessible
- Capable of inspection of packets before forwarding them to a LAN system (maybe a gateway) prior to a dedicated recipient client LAN



Firewall Diagram



Firewall Rules

- Source and Destination Addresses
 - Ports
 - Protocol
 - Action
 - Direction
-
- Rules help monitoring and filtering of packets both ingress and egress traffic

Action	Source	Destination	Protocol	Port	Direction
Allow	192.168.1.0/24	Any	TCP	80	Outbound



How does it work?

- Generally speaking, a firewall implemented within a network is responsible for enforcement of access to other portions of a network
- Prioritize rules at the top for those traffic types you will have more frequently over a network
- Firewalls process rules from **top to bottom**
- Implicitly firewalls are set to deny any-any, must remember to validate this
- Rules are written into the interface
- Firewalls can have multiple interfaces, just like a router



Linux Firewall

- Netfilter-based
 - **iptables**: Provides various functionalities to control network traffic
 - **nftables**: A successor to the “iptables” utility, with enhanced packet filtering and NAT capabilities
 - **firewalld**: Has predefined rule sets, works a bit differently from the others and comes with different pre-built network zone configurations.
- Uncomplicated Firewall (**ufw**)
 - Enable UFW: `sudo ufw enable`
 - Disable UFW: `sudo ufw disable`
 - Reset UFW rules: `sudo ufw reset`
 - Block a specific port: `sudo ufw deny 80`



Pfsense

- A free open-source firewall and router software
- Based on FreeBSD
- Can prioritize different types of traffic, top (high) to bottom (low)
- Walks through a step by step process
- You care about WAN and LAN for public facing and private facing networks (your 192s, 172s, 10s, etc)

Firewall / Rules / LAN

Floating WAN LAN GUESTNET

Rules (Drag to Change Order)

☐	States	Protocol	Source	Port	Destination	Port	Gateway	Queue	Schedule	Description	Actions
☐	✓ 3 / 4.44 MiB	IPv4 TCP	AdminPC	*	LAN address	AdministrativePorts	*	none		Allow Administrative Access	             
☐	✓ 3 / 643 KiB	IPv4 TCP/UDP	LAN net	*	LAN address	53 (DNS)	*	none		Allow DNS	              
☐	✗ 0 / 780 B	IPv4 *	LAN net	*	LAN address	*	*	none		Block Firewall Access	              
☐	✓ 27 / 331.25 MiB	IPv4 *	LAN net	*	*	*	*	none		Default allow LAN to any rule	              
☐	✓ 0 / 0 B	IPv6 *	LAN net	*	*	*	*	none		Default allow LAN IPv6 to any rule	              

 Add  Add  Delete  Save  Separator



NGFWs: GUI and CLI Accessible

```
#=====
# AUDIT COMMANDS
#=====

show admins all
show running security-policy
show running nat-policy
show rulebase security rules "Deny-All-Inbound"
show rulebase security rules "Deny-All-Outbound"
```



Palo Alto Networks

	NAME	TAGS	TYPE	Source				Destination			APPLICATION	SERVICE	ACTION	PROFILE	OPTIONS	HIT COUNT
				ZONE	ADDRESS	USER	DEVICE	ZONE	ADDRESS	DEVICE						
1	Block-QUIC-L4	none	universal	any	any	any	any	any	any	any	any	udp-80 udp-443	Deny			0
2	Block-QUIC-L7	none	universal	any	any	any	any	any	any	any	quic	application-default	Deny			0
3	Block-EDL-Source	none	universal	any	panw-bulletproof-ip-list panw-highrisk-ip-list panw-known-ip-list panw-torexit-ip-list	any	any	any	any	any	any	application-default	Deny			0
4	Block-EDL-Destination	none	universal	any	any	any	any	any	panw-bulletproof-ip-list panw-highrisk-ip-list panw-known-ip-list panw-torexit-ip-list	any	any	application-default	Deny			0
5	Intrazone-GlobalProtect	none	universal	Untrust	BE DE NL	any	any	Untrust	192.168.0.222	any	panos-global-protect	application-default	Allow			0
6	Intrazone-untrust	none	intrazone	Untrust	any	any	any	(intrazone)	any	any	any	application-default	Deny			103
7	intrazone-default		intrazone	any	any	any	any	(intrazone)	any	any	any	any	Allow	none	none	15
8	interzone-default		interzone	any	any	any	any	any	any	any	any	any	Deny	none		577



Cisco FTD

Identity Policy: [None](#)

SSL Policy: [None](#)

[Inheritance Settings](#) | [Policy Assignments \(2\)](#)

Rules | [Security Intelligence](#) | [HTTP Responses](#) | [Advanced](#)

Filter by Device

Add Category

Add Rule

Search Rules



#	Name	Source Zones	Dest Zones	Source Networks	Dest Networks	VLAN Tags	Users	Applications	Src Ports	Dest Ports	URLs	Action				
▼ Mandatory - DonotTouch (1-2)																
1	Interfaces	Internal	Internal	any	any	any	any	any	any	any	any	✓ Allow				0
2	Allow	any	any	any	any	any	any	any	any	any	any	✓ Allow				0
▼ Default - DonotTouch (-)																

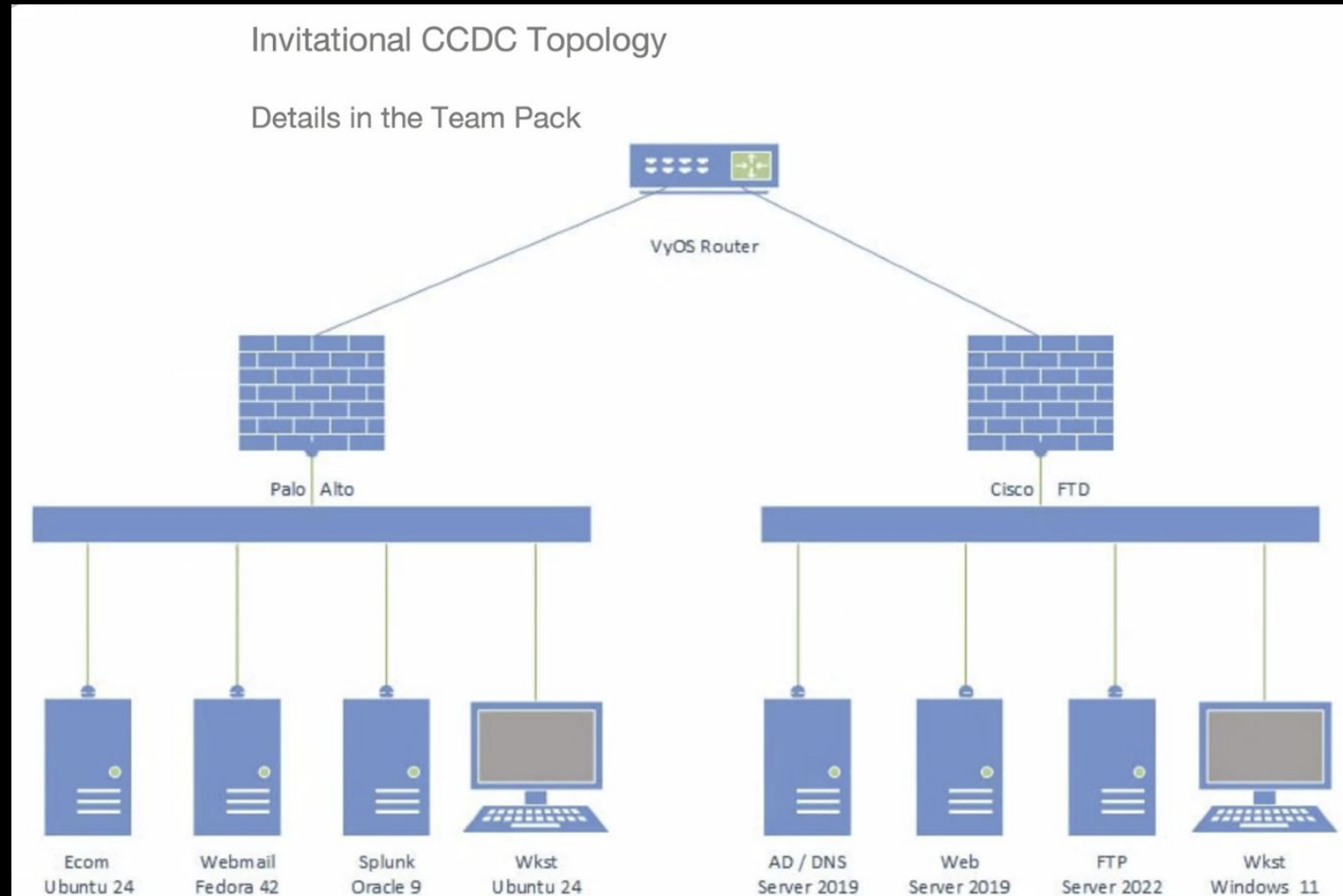
There are no rules in this section. [Add Rule](#) or [Add Category](#)

Default Action

Intrusion Prevention: Balanced Security and Connectivity



Network Topology: CCDC Midwest



Topology: John Ford Defender Toolkit

Edge

- VyOS Router

Palo Alto segment (Linux)

- Ecom: Ubuntu 24
- Webmail: Fedora 42
- Splunk: Oracle 9
- Workstation: Ubuntu 24

Cisco FTD segment (Windows)

- AD / DNS: Server 2019
- Web: Server 2019
- FTP: Server 2022
- Workstation: Windows 11

● Edge

- VyOS Router

Palo Alto segment (Linux)

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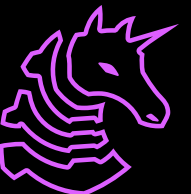
Useful Repository: John Ford's CCDC Defender Toolkit

<https://github.com/RedefiningReality/MWCCDC-Defender-Toolkit>



ELK Stack

- Popular open-source platform for collecting, searching and visualizing logs, made up of three tools:
 - **Elasticsearch** stores and indexes the data
 - **Logstash** ingests and parses it
 - **Kibana** provides dashboards and search



ELK



Elastic Search



Logstash



Beats



Kibana



Elastic: Expanding Entries

▼ March 14th 2018, 12:33:02.000

type: http **status:** Error **client_ip:** 92.108.82.117 **http.content_length:** 227
http.phrase: FOUND **http.code:** 302 **http.response_headers.content_type:** text/html
; charset=utf-8 **http.request_headers.host:** packetbeat.com **timestamp:** February 10
th 2014, 11:26:22.599 **@timestamp:** March 14th 2018, 12:33:02.000 **client_port:** 59,
902 **query:** POST /login HTTP/1.1 **path:** /login **server:** nginx-proxy2 **response:** H

Table JSON [View surrounding documents](#) [View single document](#)

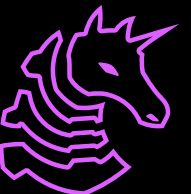
@timestamp	March 14th 2018, 12:33:02.000
_id	VHM-IWIB_763GBCM6I5v
_index	packetbeat-2018.03.14
_score	-
_type	doc
bytes_out	227
client_ip	92.108.82.117
client_location	52.366600, 4.902700
client_port	59,902
client_server	other



Lab: Splunk Basics, Splunk Querying

We need to get you to develop investigative skills, and especially familiarity with a SIEM tool. You can then be proficient for DFIR and SOC. We would like for you to eventually graduate towards using the ELK stack but the best gateway to readiness is Splunk.

<https://tryhackme.com/room/splunkexploringspl>



Additional CCDC Commentary & FAQs

<https://github.com/guyinatuxedo/dawn-guard>

<https://4rji.com/ccdc-journey>

<https://howtowinccdc.com/#/>

<https://alexlevinson.wordpress.com/2017/05/09/know-your-opponent-my-ccdc-toolbox/>

<https://malcomvetter.medium.com/how-to-win-at-ccdc-b355e309a05c>

<https://www.winterknight.net/how-to-win-ccdc-red-team/>



Signup for CCDC Invitationals

CHECK DISCORD → PURPLE ANNOUNCEMENTS



Next Meetings

2026-10-08 • This Thursday

- Windows Privilege Escalation

2026-10-13 • Next Tuesday

- TBD



ctf.sigpwny.com

sigpwny{splunkin_ccdc}

Meeting content can be found at
sigpwny.com/meetings.

